

Viral Exanthems

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Session Objectives

- Discriminate between the clinical presentations of typical viral exanthems based on exam findings and associated symptoms.
- Understand complications of typical viral exanthems and populations at higher risk.
- Discriminate between the diagnostic, treatment and prevention measures of typical viral exanthems.

Definitions

- Exanthem: rash that appear abruptly and affects several areas of the skin simultaneously
- Enanthem: rash that appears abruptly on the mucous membranes
- Morbilliform: resembling a measles rash – erythematous macules and papules

Clinical Considerations

- Most are benign
- Can be devastating for certain populations
 - Immunocompromised, unvaccinated, fetus/newborn
- Treatment is generally supportive, but targeted anti-viral treatment is available for some
- Diagnosis is primarily clinical, but specific testing is available in certain circumstances

History

- When did it start?
- Where did it start?
- How did it spread?
 - Centrifugal spread
 - Centripetal spread
 - Cephalic to caudal spread
- Evolution of lesions?
- Any inciting factors?
- Timing in relation to other symptoms
 - Prodromal/associated symptoms

Viral Exanthems

- Rubella
- Roseola
- Erythema Infectiosum
- Measles
- Chicken Pox (Varicella)
- Hand, Foot, and Mouth Disease
- Herpes Simplex
- Pityriasis rosea
- Molluscum contagiosum
- Monkeypox

Case #1

- A 2 yo male was recently diagnosed with a viral illness by his pediatrician after presenting with fatigue, runny nose, fever and rash. The rash started on the cheeks and then spread to trunk and extremities.
- Today, his 15 yo sibling, who has sickle cell anemia, presents with increased pallor and fatigue. Exam is notable for tachycardia and a systolic ejection murmur.
- Mom has a picture of the toddler's rash.



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have a hemoglobinopathy are at elevated risk for complications.
Many erythematous papules on the face, trunk, and extremities.

Which of the following viruses is most likely responsible for both children's symptoms?

- A. Measles
- B. Human Herpes Virus 6
- C. Herpes Simplex Virus 1
- D. Parvovirus B19
- E. Coxsackie Virus A

Erythema Infectiosum (5th Disease)

- Prodrome: mild “flu-like” illness [2-5 days]
 - Fever, malaise, myalgia, headache, coryza, diarrhea, abdominal pain
- Exanthem
 - Macular erythema of cheeks (“Slapped cheeks”) [2-4 days]
 - Sparing nasal, perioral, and periorbital areas
 - Facial rash fades and a pruritic reticular rash appears symmetrically on trunk and extremities [duration varies]
- No longer contagious once rash appears
- Treatment: supportive
- Diagnosis: clinical

Parvovirus B19: Other Clinical Presentations

- Symmetric small joint polyarthropathy
 - More common in adults
- Transient aplastic crisis
 - Occurs in patients with hemoglobinopathies (sickle cell), hemolytic anemias
- Chronic anemia
 - Occurs in immunosuppressed or HIV infected patients
- Hydrops fetalis
- Papular-purpuric gloves-and-socks syndrome (PPGSS)

Papular-purpuric gloves-and-socks syndrome (PPGSS)

- Adolescent girls, young adults
- Itching, burning, bright red papular exanthem on hands and feet
- Sharp border
- Edema can limit movement
- Can also see petechiae, purpura, oral vesicles
- Can also have fever, lymphadenopathy, arthralgias

Gloves and socks syndrome secondary to parvovirus B19 infection



Papular-purpuric rash on the hands and feet of a patient with acute parvovirus B19 infection.

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Exanthem occurs in 15-20% of patients, most frequent in 4-10 yo



Diffuse erythema on the face, sparing the perioral and perinasal areas.

Widespread thin, erythematous papules and plaques on the leg.



A close-up of figurate erythematous papules and plaques on an extremity.

Case #2

- A 9 mo male presents to your office with fever of 102.5°F for 3 days. He has decreased appetite and increased fussiness. Exam is notable for mild cervical LAD. No treatment is recommended other than supportive care.
- The mother calls the next day and tells you the fever is gone but her son now has a pink rash all over his body. He is otherwise doing well (eating, drinking, playful).



Discrete and confluent reddish macules, papules, and thin plaques on the arm and chest.

Which of the following is the most likely etiology of this patient's symptoms?

- A. Coxsackie virus A
- B. Human Herpes Virus 6
- C. Measles virus
- D. Parvovirus B19
- E. Varicella-zoster virus

Roseola

- High fever followed by a maculopapular rash that presents as the fever resolves
- Begins on trunk and spreads to extremities/neck/face - centrifugal
- Treatment: supportive
- Diagnosis: clinical
- Transmission: direct contact with saliva

HHV6: Other Clinical Presentations

- Can remain latent and reactivate later in life
- Can cause meningitis, encephalitis in immunocompromised patients
- In patients who have undergone hematopoietic stem cell transplant ganciclovir and foscarnet may be beneficial
- Testing in these more extreme cases should be done with PCR
 - Blood, CSF, tissue



upper respiratory mucosa is about 10 days (range 7-21 days), with the rash typically appearing about 14 days after onset of symptoms.
 Widespread discrete and confluent erythematous macules and patches on the trunk.



Following brief gentle pressure with a fingertip or microscope slide to blanch involved skin may reveal underlying pinpoint hemorrhages.
 Discrete and confluent rose-pink macules, papules, and thin plaques on the cheek.

Case #3

- Female infant born full term via SVD to a mother who recently immigrated from India. No prenatal records are available. During the initial newborn exam the physician notes the infant to be small for gestational age, a heart murmur, hepatosplenomegaly and a papular bluish rash.



Zitelli and Davis' Atlas of Pediatric Physical Diagnosis, 7th edition, 2018. Chapter 13 Infection Diseases, Marian G. Michaels and John V. Williams

The illness in this infant could have been prevented by which of the following measures?

- A. Immunization of the mother prior to pregnancy
- B. Immunization of the mother during pregnancy
- C. Administration of an antiviral medication to the mother during pregnancy
- D. Isolation of the infant immediately after birth
- E. Delivery of the infant by C-section

Congenital Rubella Syndrome

- Cataracts, retinopathy, glaucoma
- Sensorineural hearing loss
- Patent ductus arteriosus, pulmonic stenosis
- Hepatosplenomegaly, jaundice
- Growth restriction
- Intellectual disability
- Erythematous maculopapular rash
 - Dermal erythropoiesis “Blueberry muffin”
- Diagnosis: serology (IgM, IgG), PCR (throat, nasal, urine), viral cx (throat, nasal, urine)

MMR Vaccine

- Live attenuated virus
- 2 doses
 - 12 - 15 mo
 - 4 - 6 yo
- 95% conversion after single dose
- Life-long immunity in >90%
- Contraindications
 - Altered immunity
 - Anaphylaxis
 - Pregnancy

Rubella (German Measles)

- Prodrome: mild or absent
 - Low grade fever, sore throat, malaise
- Lymphadenopathy: posterior auricular, occipital, can be generalized
- Exanthem: fine, erythematous, maculopapular rash [1-3 days]
 - Starts on face and becomes generalized in 24 hours,
 - Enanthem: petechiae on soft palate (Forscheimer spots)
- Transmission: droplet, direct contact with nasopharyngeal secretions, transplacental
- Peaks in late winter/early spring
- Most infections now occur outside US or in under-immunized individuals
- Treatment: supportive care



Widespread deeply erythematous and violaceous macules and patches on the legs.



Image source: CDC

Rubella rash on a child's back



The distribution is similar to that of measles (rubeola), though the lesions are less intensely red.

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Case #4

- An 11 mo male presents to your office with fever, generalized rash, red eyes, cough and runny nose. He is irritable and not eating. His family just returned from a trip to Eastern Europe. No one else in the family is sick. Parents first noted the rash behind the ears and along his hairline.



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s after acute infection. This complication A close-up of erythematous macules, patches, and plaques.

Which of the following infections is most likely responsible for the patient's symptoms?

- A. Varicella-zoster virus
- B. Parvovirus B19
- C. Coxsackie virus A
- D. Human herpes virus 6
- E. Measles virus

Measles

- One of the most contagious viruses (Infectious 4 days before to 4 days after onset of rash)
- Transmission: direct contact with large droplets or contact with small droplet aerosols
 - Virus can remain suspended for up to 2 hrs in the air
- 1 infectious person infects about 12-18 individuals

Measles

- Incubation: asymptomatic [10-14 days]
- Prodrome: high grade fever, cough, coryza, conjunctivitis, pharyngitis, enanthem (Koplik spots) [2-3 days]
- Exanthem: starts 1-2 days after prodrome
 - Erythematous nonpruritic macules/patches on face
 - Koplik spots start to resolve prior to facial rash
 - Spreads to the trunk and extremities within 2-3 days
 - Palms and soles involved

Measles

- Diagnosis: clinical but can confirm with testing
 - Immunofluorescence of nasal mucosal cells
 - PCR of nasal secretions, blood, urine
 - Serology: IgM, 4 fold increase in IgG
 - Viral cell culture
- Treatment: vitamin A, ribavirin
- Postexposure prophylaxis: IVIG or vaccination with MMR/MMRV within 72 hours of exposure

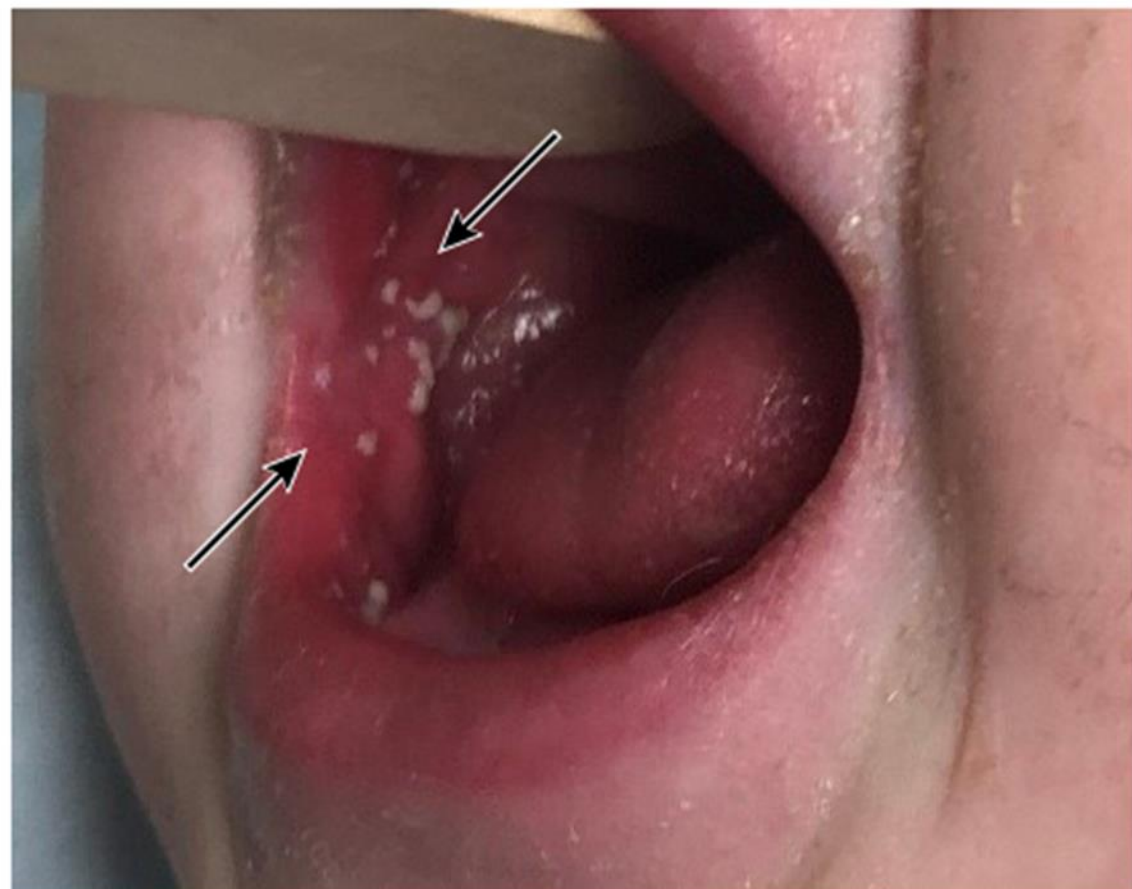
Measles Complications

- Diarrhea
- Otitis media
- Croup, bronchiolitis
- Pneumonia is the most common cause of death from measles in young children
 - Primary viral or bacterial superinfection
- Encephalitis (15% mortality, 20-40% long-term sequelae)
 - Post-infectious, immune mediated
 - Uncommon – 1 in 1,000 cases
- Myocarditis
- Subacute sclerosing panencephalitis (SSPE)
 - Late complication, rare
 - Virus remains latent in CNS and reactivates 7-10 years later
 - Attacks CNS cells, leads to cell death, neurodegenerative process

MMR Vaccine

- Live attenuated virus
- 2 doses
 - 12 - 15 mo
 - 4 - 6 yo
- 95% conversion after single dose
- Life-long immunity in >90%
- Contraindications
 - Altered immunity
 - Anaphylaxis
 - Pregnancy
- 95% of the population needs to be immunized in order to achieve herd immunity

Koplik spots



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Measles exanthem



Blanching erythematous macules with some confluent areas on the trunk in a patient with measles.

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Case #5

- A 2 yo male brought to clinic due to two days of fever, fussiness, decrease in appetite. He is up to date on immunizations. There is no focus of infection on exam. You send him home with instructions for supportive care. He returns the next day with the following:



may cause atypical HFMD, where more extensive lesions may occur on the buccal mucosa and in the perioral area. "Tomato flu"

Tiny vesicles and surrounding erythema on the buccal mucosa.

What is the most likely etiology of the patient's findings?

- A. Coxsackie virus A
- B. Varicella-zoster virus
- C. Herpes simplex virus 1
- D. Human herpes virus 6
- E. Measles virus

Hand, Foot and Mouth Disease

- Caused by enteroviruses: coxsackie viruses, enteroviruses, echoviruses
- Peaks in summer/early fall
- Transmission: fecal-oral, respiratory
- Most commonly seen in children <10 yo
- Diagnosis: clinical but can confirm with PCR (vesicle fluid, blood, urine, CSF, stool, respiratory secretions)
- Treatment: supportive

Hand, Foot and Mouth Disease

- Prodrome: absent or mild [4-6 days]
 - Vomiting, diarrhea, abdominal pain
- Low grade fever can precede enanthem and exanthem
- Enanthem: painful, erythematous macules and vesicles
 - Commonly rupture and form superficial ulcer with gray-yellow base and erythematous rim
 - Refusal to eat or decreased oral intake is common
 - If oral involvement only it is referred to as herpangina
- Exanthem: macules, papules, cloudy vesicles with erythematous base (halo), non pruritic, usually nontender
 - Occurs on the hands, palms, feet, soles, buttocks, extremities



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ly elevated, and atypical lymphocytes. Oval cloudy vesicles on a background of erythema on the heel.



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may cause atypical HFMD, where more extensive vesicle formation is seen and in the perioral area. Scattered erythematous papules and oval cloudy vesicles on the palm and fingers.

Case #6

- A 6 yo male presents with fever, cold symptoms, headache, fatigue and diffuse pruritic rash. The rash started 2 days after the fever. It was initially on the trunk and spread to all extremities. Some lesions are now crusted over. There are also some painful lesions in the mouth. He is with foster parents. They are not sure what immunizations the child has received.



Scattered papules, papulovesicles, pustules, and crusts on the thumbs.



Several small papules, vesicles (some umbilicated), and crusts on the back.

The illness in this child can typically be prevented by which of the following measures?

- A. Vaccination with a live attenuated virus vaccine
- B. Vaccination with a killed virus vaccine
- C. Vaccination with a polysaccharide vaccine
- D. Vaccination with a protein conjugate vaccine

Chicken Pox (Varicella)

- Prodrome: fever, fatigue, anorexia, congestion, rhinorrhea, cough
- Exanthem: very pruritic erythematous rash on trunk and scalp and then spreads peripherally (centrifugal)
 - Macules > papules > vesicles > pustules with crusting
 - Lesions at various stages, “Dew drop on rose petal”
- Enanthem: uncommon, painful vesicles > erosions on palate
- Transmission: airborne, direct contact of lesions, transplacental
- Diagnosis: clinical
 - Testing with PCR, DFA, culture if needed
- Treatment: supportive

Chicken Pox Complications

- Mainly in the immunocompromised
- Treat with antivirals in these patients: acyclovir, valacyclovir, famciclovir
- Bacterial superinfection (staph, strep): impetigo, sepsis, osteomyelitis
- Pneumonia: primary viral or bacterial superinfection
- Neuro: meningitis, encephalitis, Guillain-Barre Syndrome
- Reye syndrome

Varicella Vaccine

- Live attenuated virus
- Two doses
 - 12-15 mo
 - 4-6 yo
- 92% effective after two doses
- Contraindications
 - Altered immunity
 - Anaphylaxis
 - Pregnancy



Numerous tiny vesicles and crusts on the lower face.



The enanthem of chickenpox showing erosions with overlying slough on the palate.

Case #7

- An 18 yo male with history of asthma just completed a course of prednisone for treatment of a severe asthma exacerbation. He presents to your office today for follow up and states that he is having “chest pain”. He has no respiratory or other concerning symptoms. On exam you note the following:



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Grouped vesicles on an erythematous base in a T3 distribution and few scattered papules outside the dermatome.

What is the most likely cause of his symptoms?

- A. Trauma
- B. Reactivation of a latent virus
- C. A new infection that he likely acquired while hospitalized for the recent asthma exacerbation
- D. A burn

Shingles (Herpes zoster)

- Prodrome: Pain, tenderness, paresthesia, allodynia, intense itching in affected dermatome often precedes rash
- Exanthem: erythematous papules [24 hrs] > vesicles-bullae [48 hrs] > pustules [96 hrs] > crusts [7 to 10 days]
 - New lesions continue to appear for up to 1 week
 - Can affect adjacent dermatome (e.g. T4 and T5)
- Enanthem: painful erythematous vesicles and erosions can appear on mucous membranes of affected dermatome (i.e. mouth, vagina, urethra)
- Complications: postherpetic neuralgia occurs in ~40% of pts >60 yo
 - Resolution in 87% of pts at 6 mos

Ramsey Hunt Syndrome (Herpes Zoster Oticus)

- Involvement of geniculate ganglion and CNVIII
- Prodrome: Triad of 1) ipsilateral facial paralysis 2) ear pain 3) vesicles in auditory canal and auricle
- Symptoms in addition to exanthem
 - Ipsilateral altered taste, decreased hearing, tinnitus, lacrimation, vertigo
- Treatment: acyclovir, valacyclovir, famciclovir and steroids



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Varicella-Zoster virus (VZV) is the cause of shingles. Herpes zoster (shingles) generally starts with a 1- to 3-day prodrome of otalgia and ipsilateral facial
Grouped papules and crusting on the preauricular skin.

Herpes Zoster Ophthalmicus

- Involvement of ophthalmic division of CNV
- Prodrome: headache, malaise, fever
- Symptoms in addition to exanthem
 - Unilateral pain of eye, forehead and top of head
 - Conjunctivitis, uveitis, episcleritis, keratitis
- Complications: potential loss of vision (early diagnosis critical)
- Treatment: acyclovir, valacyclovir, famciclovir



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Zoster ophthalmicus displaying grouped vesicles and pustules in the V1 dermatome along with periorbital edema and hemorrhagic crusts on the lower eyelid.

Herpes Zoster Vaccine

- 50 years of age and older
- To reduce risk of developing herpes zoster
- Two types
 - Recombinant zoster vaccine (RZV)
 - Varicella-zoster virus glycoprotein E in combination with an adjuvant
 - Zoster vaccine live (ZVL)
 - Live attenuated vaccine

Case #8

- A 3 yo boy presents to clinic on Christmas Eve with refusal to eat and drink since yesterday, fever to 102.5°F and fussiness. Exam is notable for ulcers on the oral mucosa and erythematous and bleeding gums. There is no skin rash.



Zitelli and Davis' Atlas of Pediatric Physical Diagnosis, 7th edition, 2018. Chapter 13 Infection Diseases, Marian G. Michaels and John V. Williams

The patient's symptoms are most consistent with infection with which of the following organisms?

- A. Herpes simplex virus 2
- B. Herpes simplex virus 1
- C. Coxsackie virus A
- D. Varicella-zoster virus
- E. Human herpes virus 6

Herpes Simplex Virus (1 and 2)

- Typically, HSV1 causes cold sores and HSV2 causes genital herpes, **BUT**, both viruses can cause oral or genital infection (Xu J Inf Disease 2002)
- Each virus causes more severe disease if infecting it's typical location
- HSV1 prevalence: more than 25% of children will be infected by the age of 7 and more than 30% of adolescents will be infected by 19 yo (Xu J Pediatr 2007)
- Primary infection will have more systemic symptoms than recurrent infections
- Transmission: saliva, skin-skin contact through vesicular fluid, sexual

Herpes Simplex Virus Manifestations

- Oral: gingivostomatitis, labialis, pharyngitis
- Skin
 - Herpetic whitlow: infection of the finger
 - In children, can occur at the time of primary oral infection through autoinoculation
 - Herpetic gladiatorum: infection of face, neck and arms; most commonly in wrestlers
 - Eczema herpeticum: hx of atopic dermatitis where HSV causes a secondary infection
 - Erythema multiforme
- Ocular: keratoconjunctivitis, chorioretinitis (neonates)
- Diagnosis: clinical or obtain swab for PCR from the vesicular fluid or ulcer

HSV Recurrence

- Reactivation triggers: sun, menses, trauma, decreased immunity
- Tend to be milder than initial presentation in immunocompetent
- Can be severe in immunosuppressed: disseminated disease, encephalitis, meningitis, keratitis
- Treatment: supportive, acyclovir, valacyclovir, famciclovir as needed
 - IV antivirals for neonates, encephalitis, immunocompromised
 - Can use antivirals daily for prophylaxis to reduce severity, frequency, duration of recurrent infections (HSV1 recurrence typically less than HSV2 recurrence; 1x/yr vs 4x/yr on avg)



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Grouped vesicles, pustules, and crusting on the lower cutaneous lip.

Case #9

- An 18 yo male presents with chief complaint of “itchy rash”. Rash started one week ago with a single spot but has spread to his back, abdomen and chest. He has no other symptoms. He takes no medications. He has not spent a significant amount of time outside recently.



Multiple scaly erythematous papules and oval plaques over the trunk and upper arms. Note the herald patch on the left abdomen.

What is the most likely diagnosis?

- A. Pityriasis rosea
- B. Eczema
- C. An allergic reaction
- D. Tinea corporis
- E. Scabies

Pityriasis Rosea

- Exanthem:
 - Herald patch: 3-5cm isolated, oval, scaly pink patch with central clearing
 - Over 1-2 weeks will develop very pruritic, scaly, erythematous plaques, with colarette of scale in Christmas tree pattern (spreads cephalo-caudal)
- Diagnosis: clinical
 - Testing to exclude tinea or syphilis may be considered
- Treatment: supportive (antihistamine, topical steroid, UV light)
- Will resolve spontaneously in 6-8 wks



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An oval plaque with central fine white scale ("herald patch") on the upper right chest and smaller similar plaques in the vicinity.



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Large ovoid, scaly, erythematous, and hypopigmented plaques on the back.

Case #10

- An 18 mo male brought to clinic for a well child visit. His mother's only concern is a rash on his neck that has been present for two months and seems to be getting worse. The rash does not bother him at all. He has no other symptoms. He takes no medications.



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Smooth white papules, some with secondary excoriation and inflammation, on the cheek and upper neck.

What would you tell the mother to expect regarding the rash?

- A. The rash will resolve on its own in 1-2 weeks
- B. The rash will require treatment with acyclovir
- C. The rash will resolve on its own over the next several months
- D. The rash will require treatment by a dermatologist
- E. The rash will need to be biopsied

Molluscum

- Exanthem: discrete flesh-colored domed papules with central umbilication (2-5 mm)
 - Common on trunk, extremities, face
 - Asymptomatic generally but may be pruritic
 - Can get local eczematous reaction
- Transmission: skin to skin contact, autoinoculation through scratching
- Treatment: curettage, topicals, light therapy (cosmetic)
- Commonly resolves spontaneously over 2-24 months



A close-up of numerous reddish, dome-shaped, umbilicated papules of varying sizes.



Infected molluscum contagiosum showing a cluster of crusted and purulent papules with more typical papules (smooth, skin-colored, and whitish) superiorly.

Case #11

- A 7-year-old boy is brought to the pediatric clinic by his mother with a 3-day history of fever, fatigue, and a spreading rash. The rash initially appeared as small red spots on his face but has since progressed to involve his arms, legs, and trunk. Exam reveals erythematous pustules. Mom reports the child is up to date on vaccinations including varicella. The family recently returned from a Mediterranean cruise.



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high risk for mpox 2003 outbreak: A close-up of an umbilicated and crusted pustule with an erythematous border.

What is the most likely etiology of the patient's symptoms?

- A. Chicken pox
- B. Hand-Foot-Mouth Disease
- C. Molluscum
- D. Monkeypox
- E. Epstein-Barr Virus

Monkeypox (Mpox)

- Viral zoonotic infection caused by monkeypox virus (MPXV)
- Transmission: skin-to-skin or mucosal contact (fomite possibly)
- Prodrome: fever, chills, myalgias, malaise [1-5 days]
- Exanthem: develops 1-3 days after fever and starts on the face and extremities (including palms and soles) with centripetal spread
 - Macules [1-2 days] > papules [1-2 days] > vesicles [1-2 days] > pustules [5-7 days] > crusts [up to 14 days]
 - Lesions in any one area of the body will be at the same stage and progress simultaneously

Mpox Treatment and Vaccine

- Treatment: supportive, most recover without any medical intervention.
 - Tecoviromat: originally developed to treat smallpox, used during outbreak under an Expanded Access Investigational New Drug protocol in the US
- Two vaccines available that may reduce risk of developing Mpox
 - MVA: made from a highly attenuated, nonreplicating vaccinia virus.
 - Approved in US for smallpox and monkeypox
 - ACAM2000: replication-competent smallpox vaccine
 - Only used in select patients and is associated with more adverse events
 - Only approved in US for the prevention of smallpox
 - Can be used for the prevention of monkeypox under an expanded-access investigational new drug (IND)



1996-97 outbreak: Umbilicated vesicles and crusted lesions scattered over the trunk and arm.

Summary Slide

- Specific items in the history are very important for a patient presenting with a rash – timing, evolution, spread
- Describe the rash using terminology of primary and secondary lesions
- Viral exanthems can have similar appearances, but the history, distribution, and evolution of the rash can help with the differential diagnosis
- Viral exanthems can often be diagnosed clinically, but confirmatory testing is used in certain situations
- Treatment for viral exanthems is often supportive, but some antiviral agents are available and are used in special situations such as in immunocompromised patients
- While many viral exanthems are self-limited and mild infections, certain populations are at risk for more severe infection and complications

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Answers

1. D
2. B
3. A
4. E
5. A
6. A
7. B
8. B
9. A
10. C
11. D